COVID-19’s Impact on Motor Vehicle Collisions in NYC

STAT 605 Final Presentation

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Introduction

Project Overview

Research Question: How did COVID-19 affect motor vehicle collision patterns in New York City?

- **Dataset:** 2.22M police-reported crashes (2012-2025)
- **Time Periods:**
  - Pre-COVID (before March 2020)
  - During COVID (March 2020 - Dec 2021)
  - Post-COVID (2022-present)
- **Key Questions:**
  1. Did collision rates return to pre-pandemic levels?
  2. Did crashes become more deadly?
  3. What changed in driver behavior?

Data Sources

Datasets:

- **Primary:** NYC Motor Vehicle Collisions (2.22M records)
- **Secondary:** Vehicle-Level Details (4.45M records)
- **Tertiary:** NYC Speed Limits (141K segments)
- **Quaternary:** Twitter Traffic Reports (51K tweets)

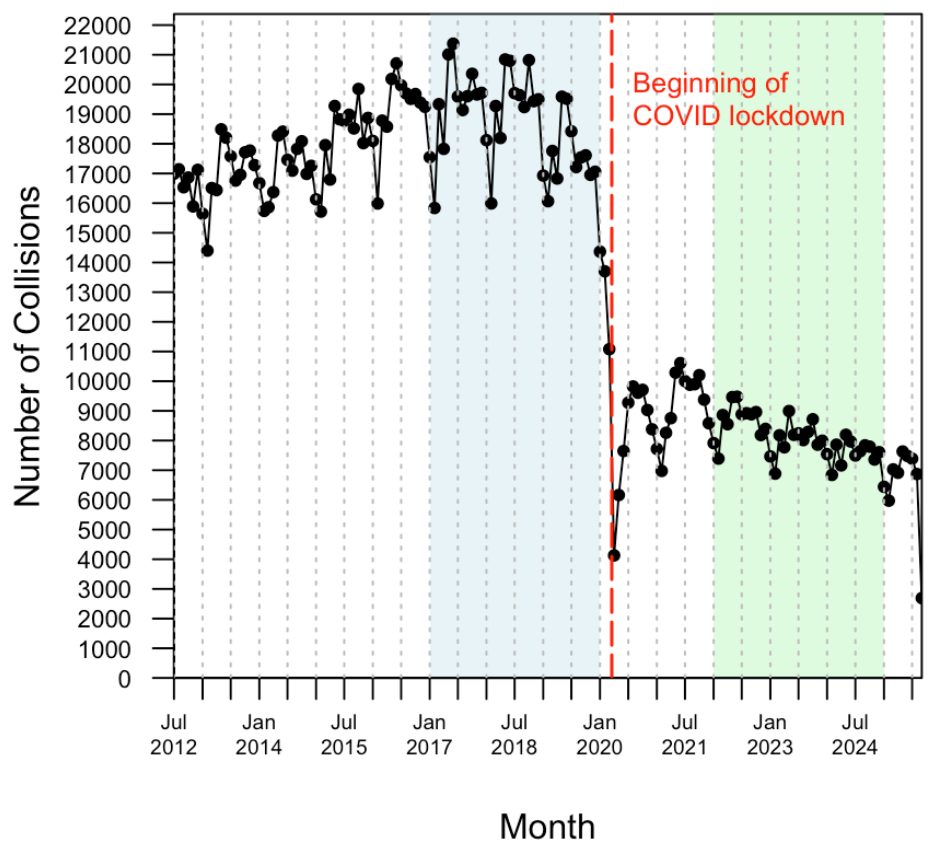
Key Variables:

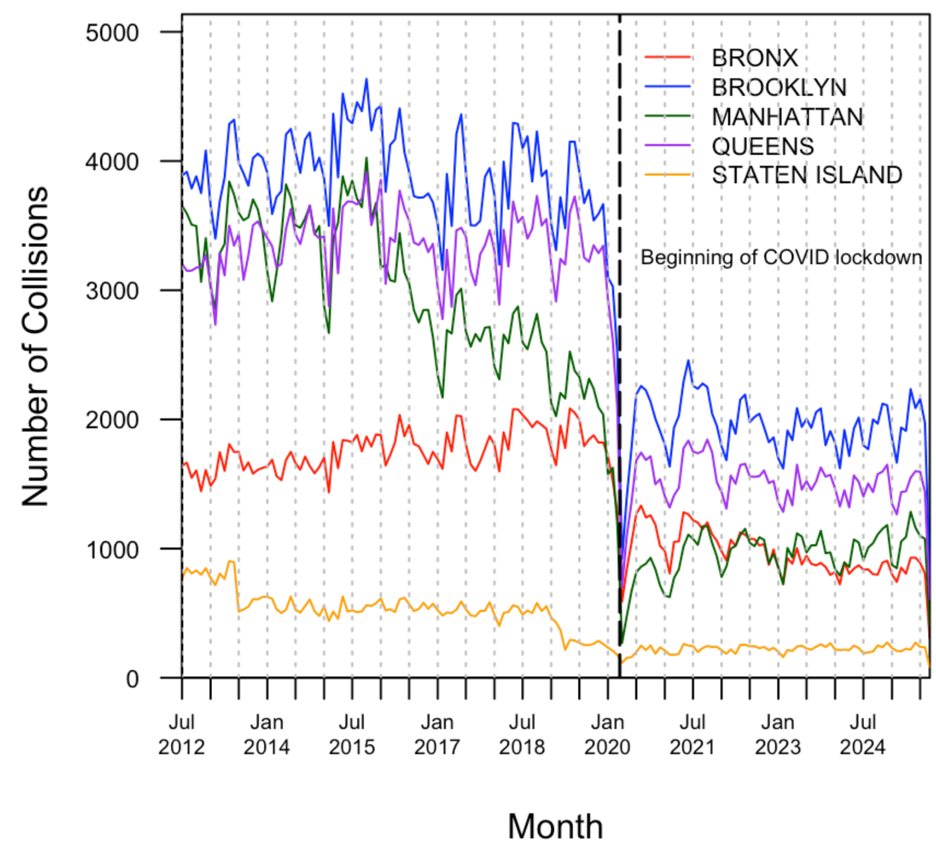
- Crash date/time & location
- Casualties by road user type
- Contributing factors (up to 5)
- Vehicle characteristics
- Borough information

Did Collision Rates Change?

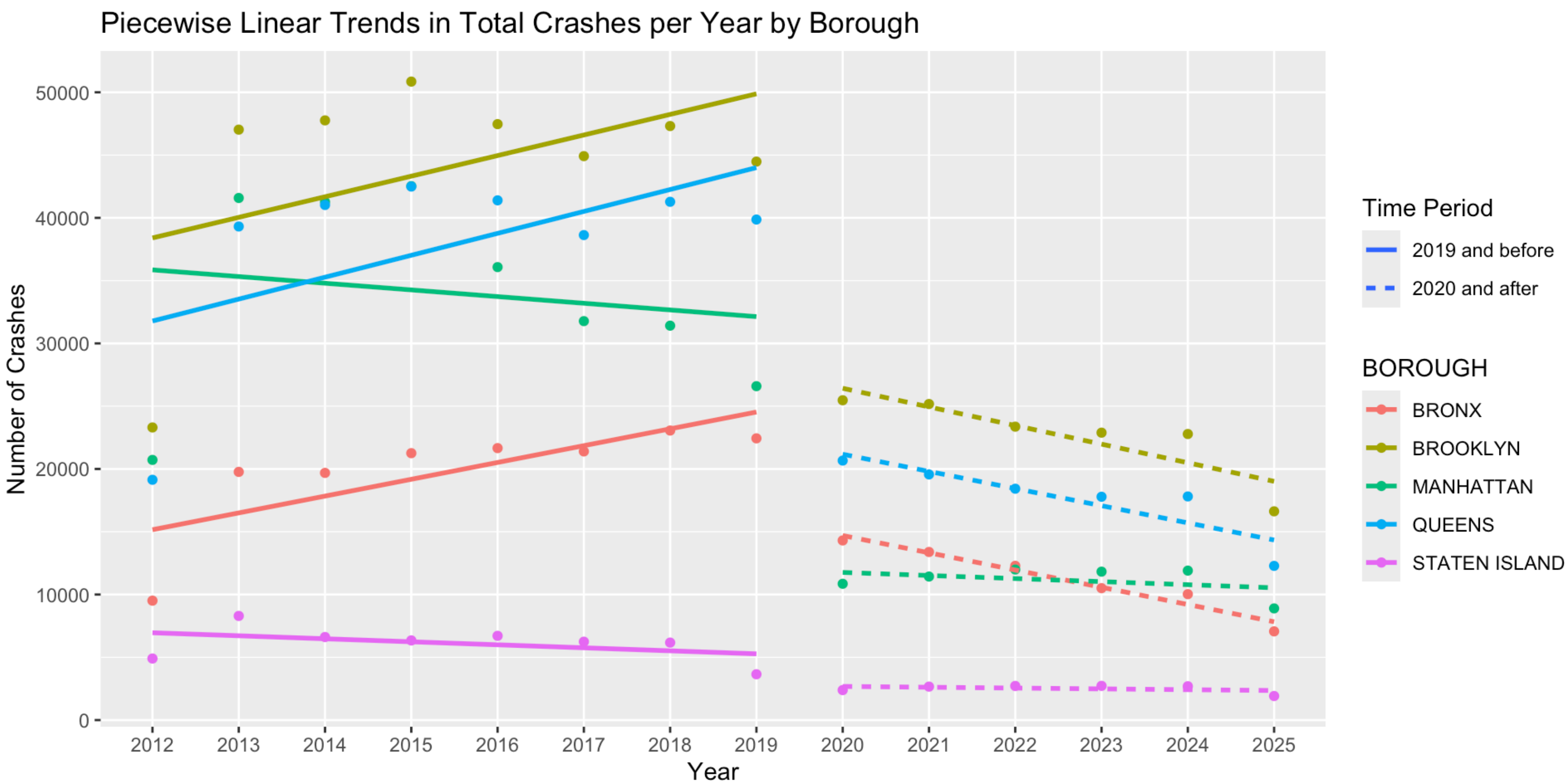
Monthly Collision Trends

Number of Monthly Vehicle Collisions in NYC:





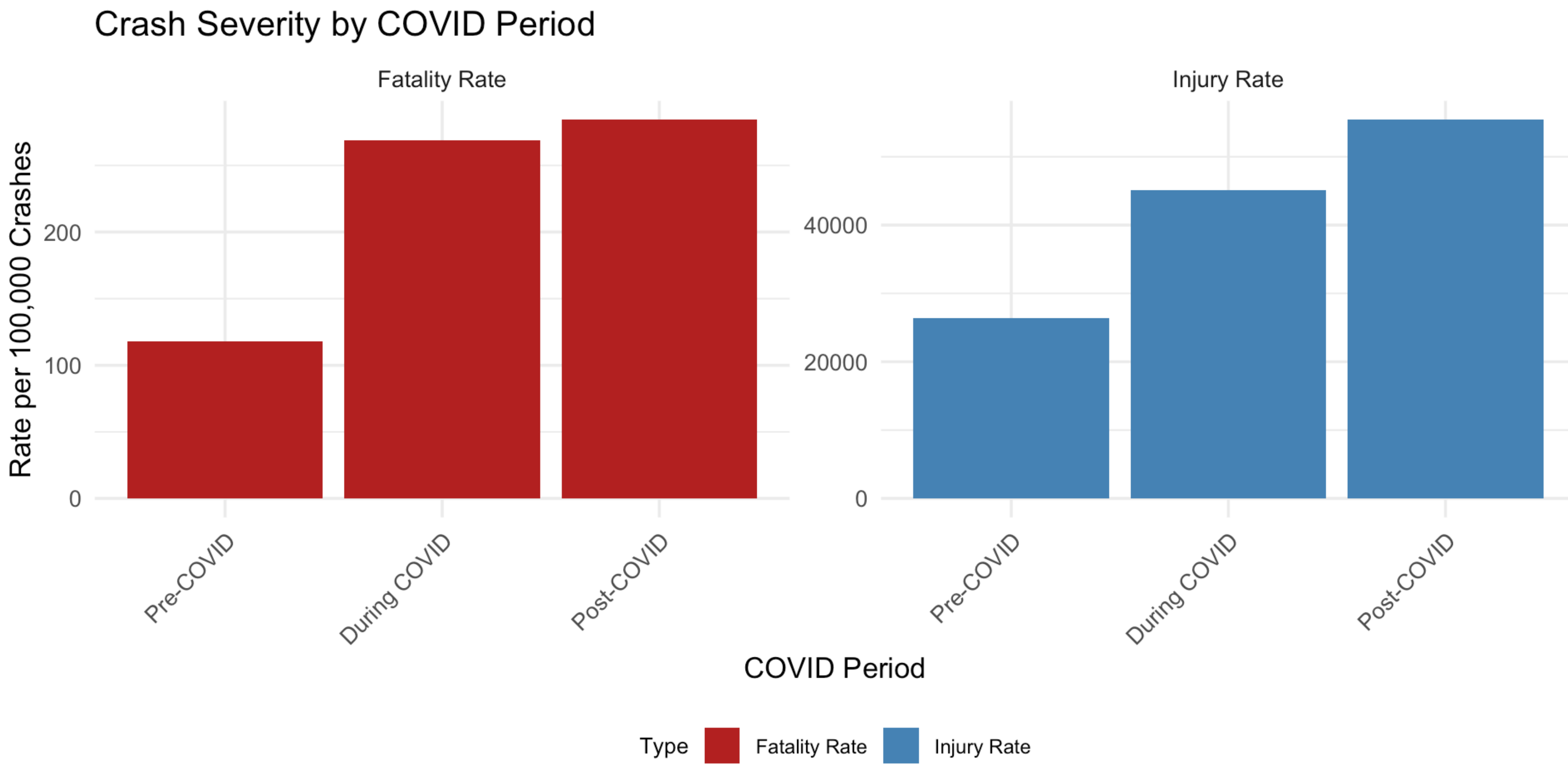
Borough-Level Patterns



- **Brooklyn & Queens & Bronx:** Increasing pre-COVID, Decreasing post-COVID
- **Manhattan:** Decreasing pre-COVID, more static post-COVID

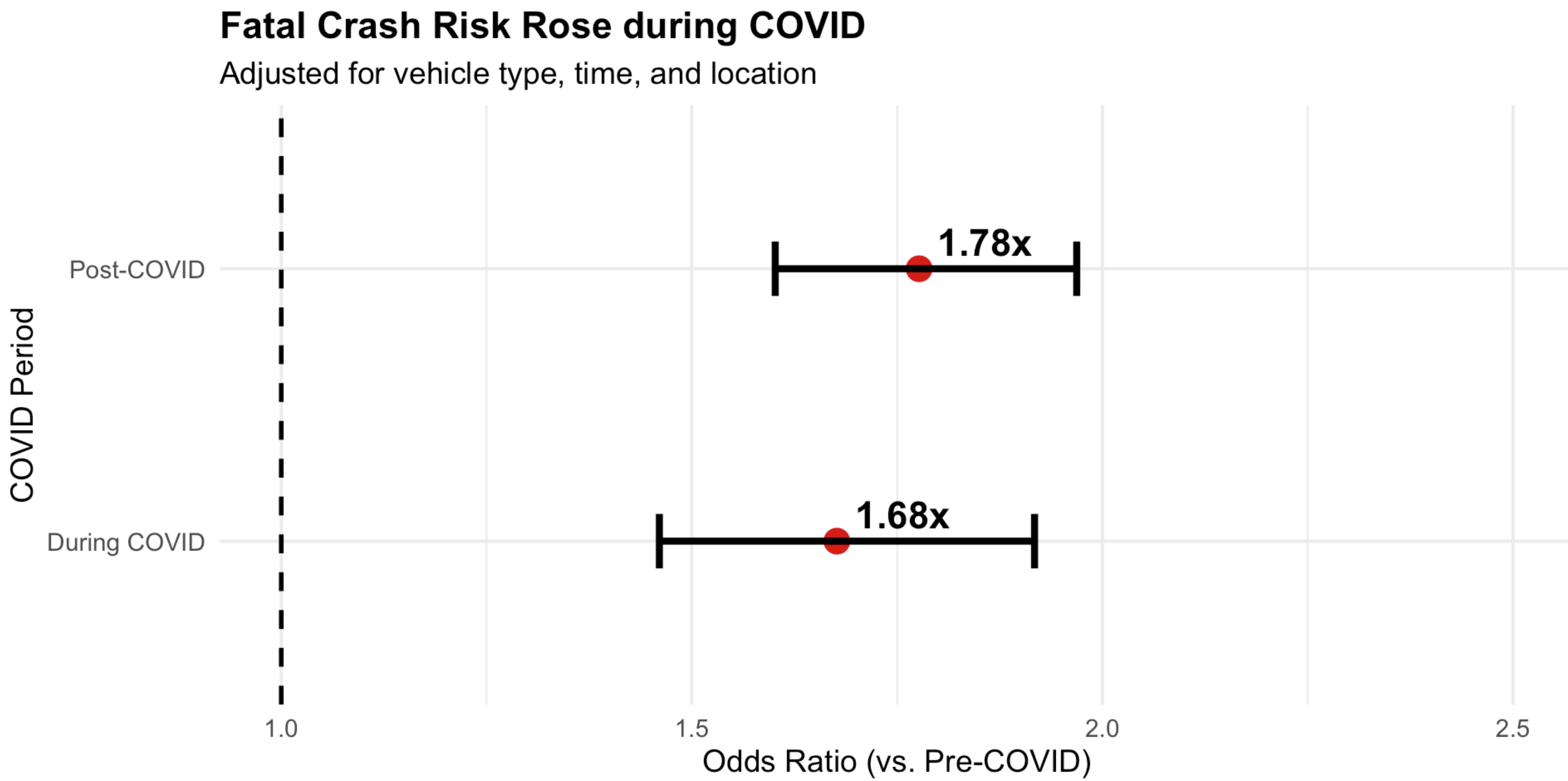
Did Crashes Become More Severe?

Severity



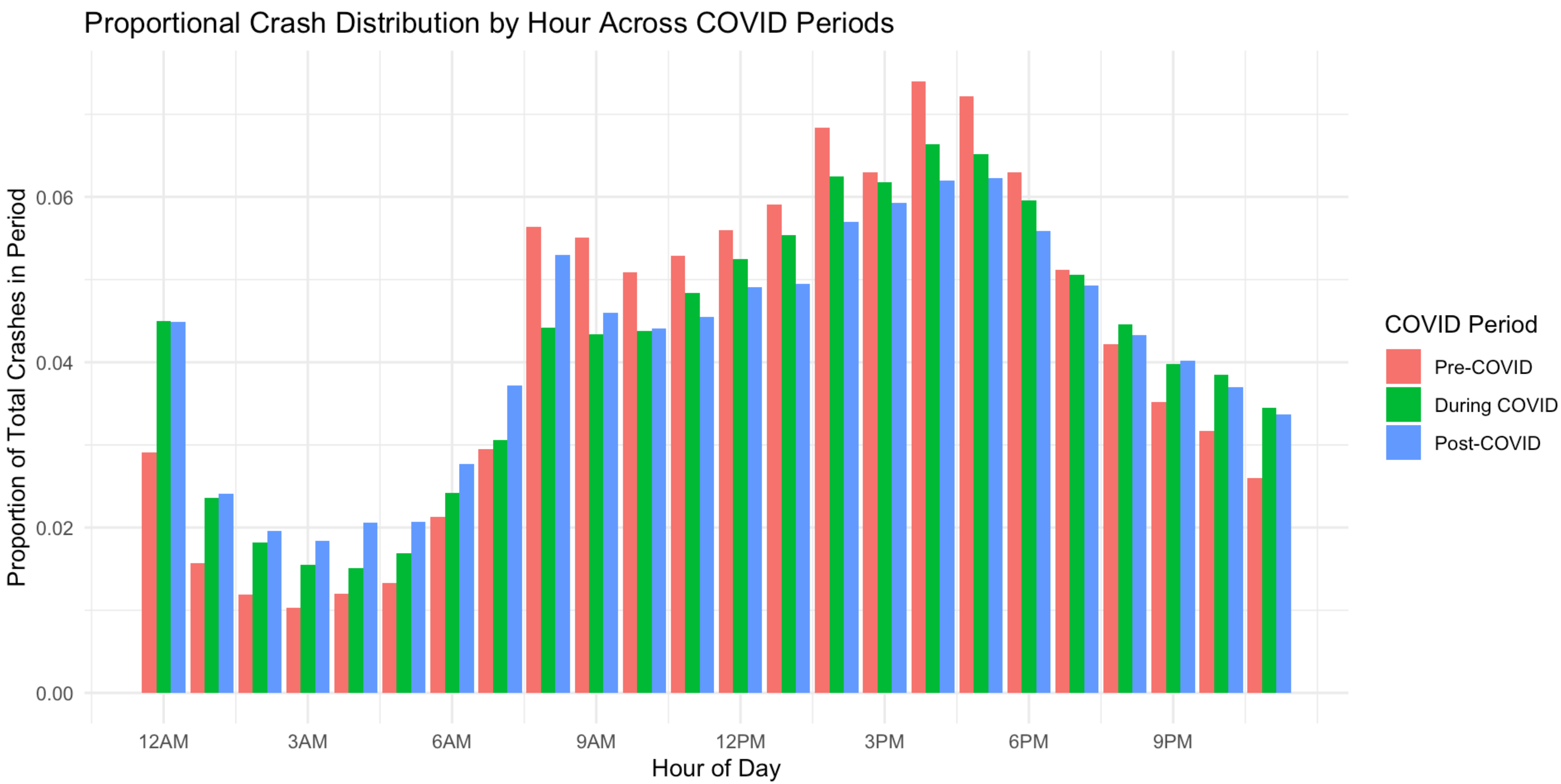
Key Finding: Crash severity increased during and after COVID-19

COVID's Impact on Severity



Key Finding: Even controlling for other factors, crashes became 67-78% more likely to be fatal

Crash Timing Patterns

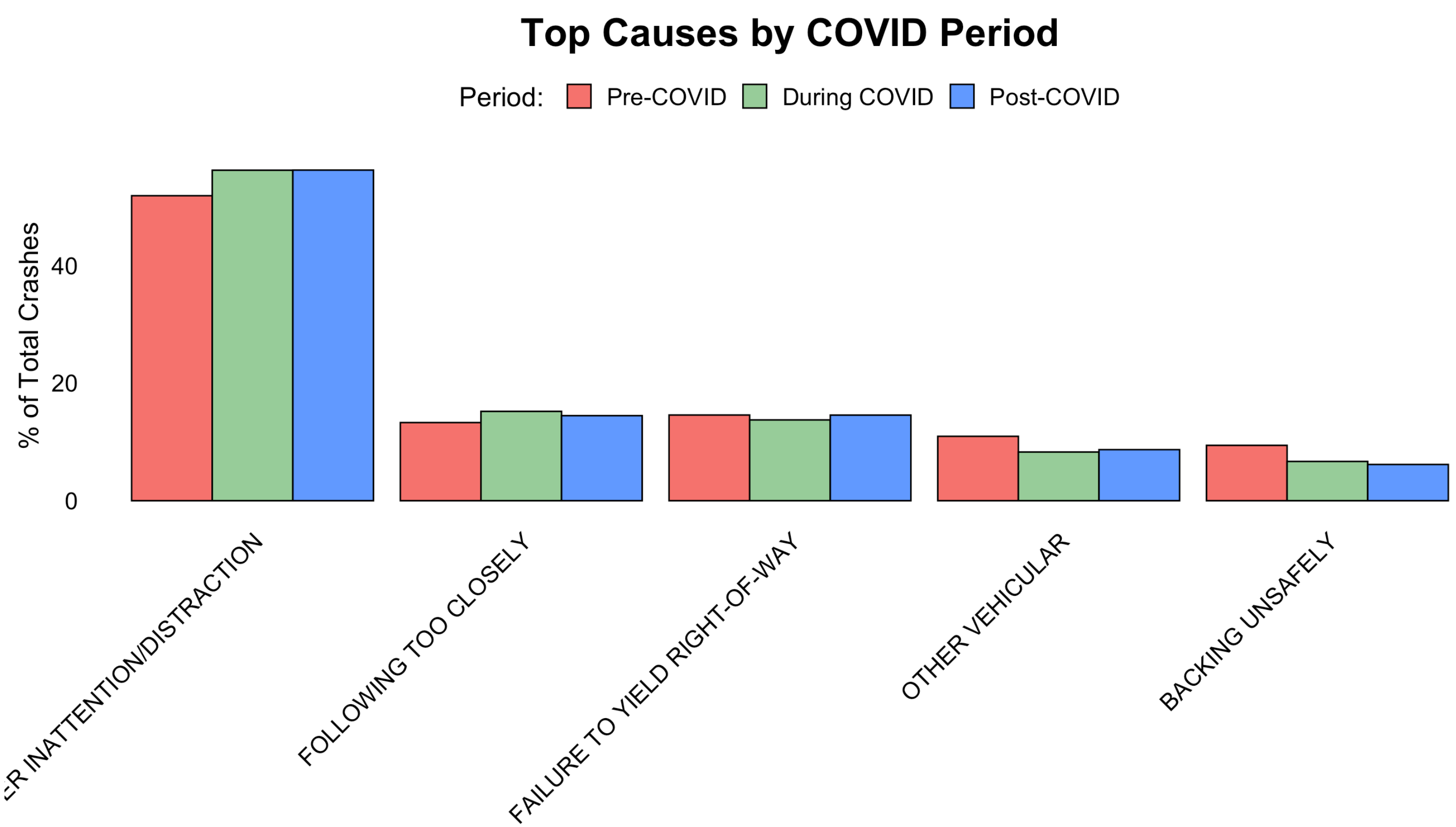


Rush-hour peaks dipped slightly during COVID but largely returned afterward

What Caused These Changes?

Contributing Factors

How Factors Changed:



Key Finding: Driver inattention and distraction are the leading cause and became more common during COVID

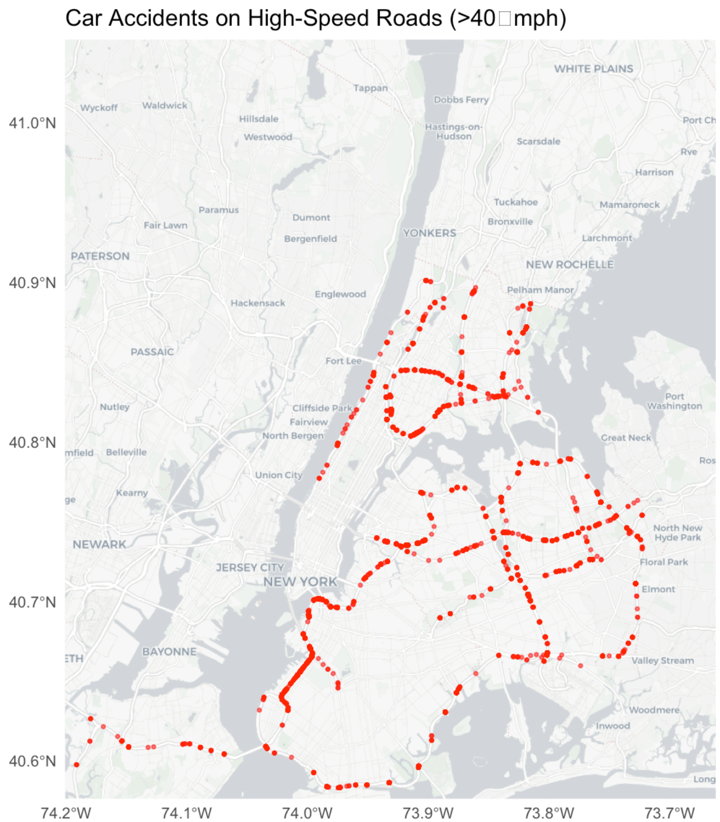
Text Mining Analysis

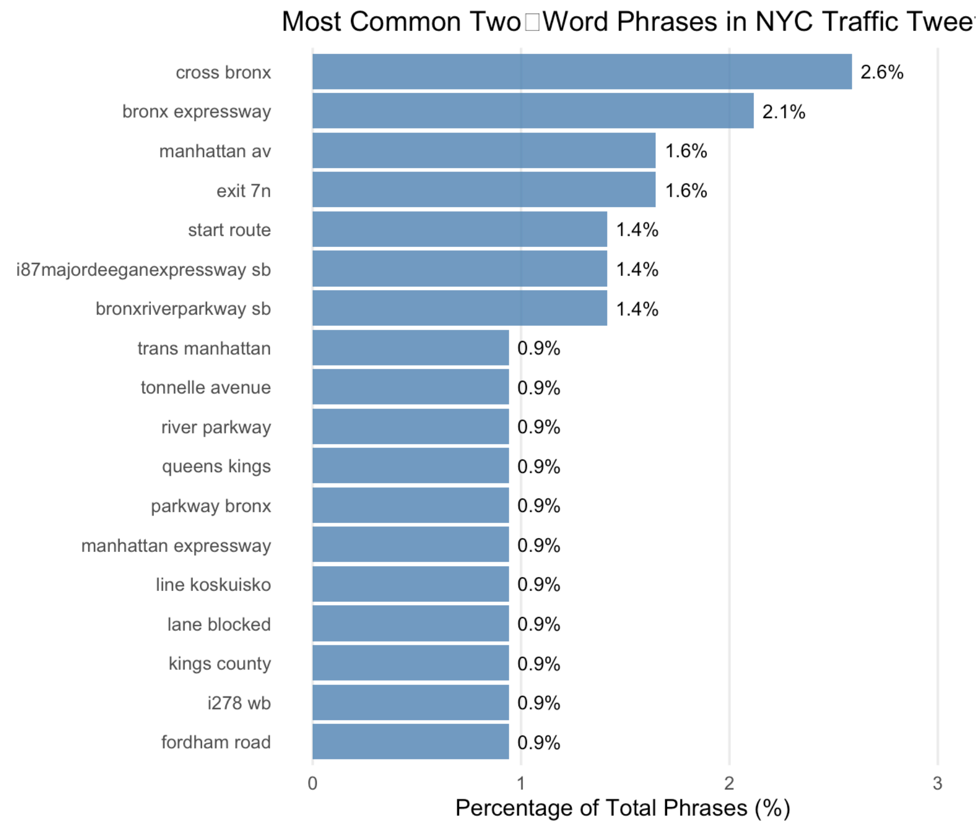
Word Cloud

Data Processing:

- Started with 51K traffic-related tweets
- Filtered to NYC-specific mentions
- Isolated vehicle crashes (excluded transit)
- Final dataset: **700+ vehicle crash tweets from 2018**

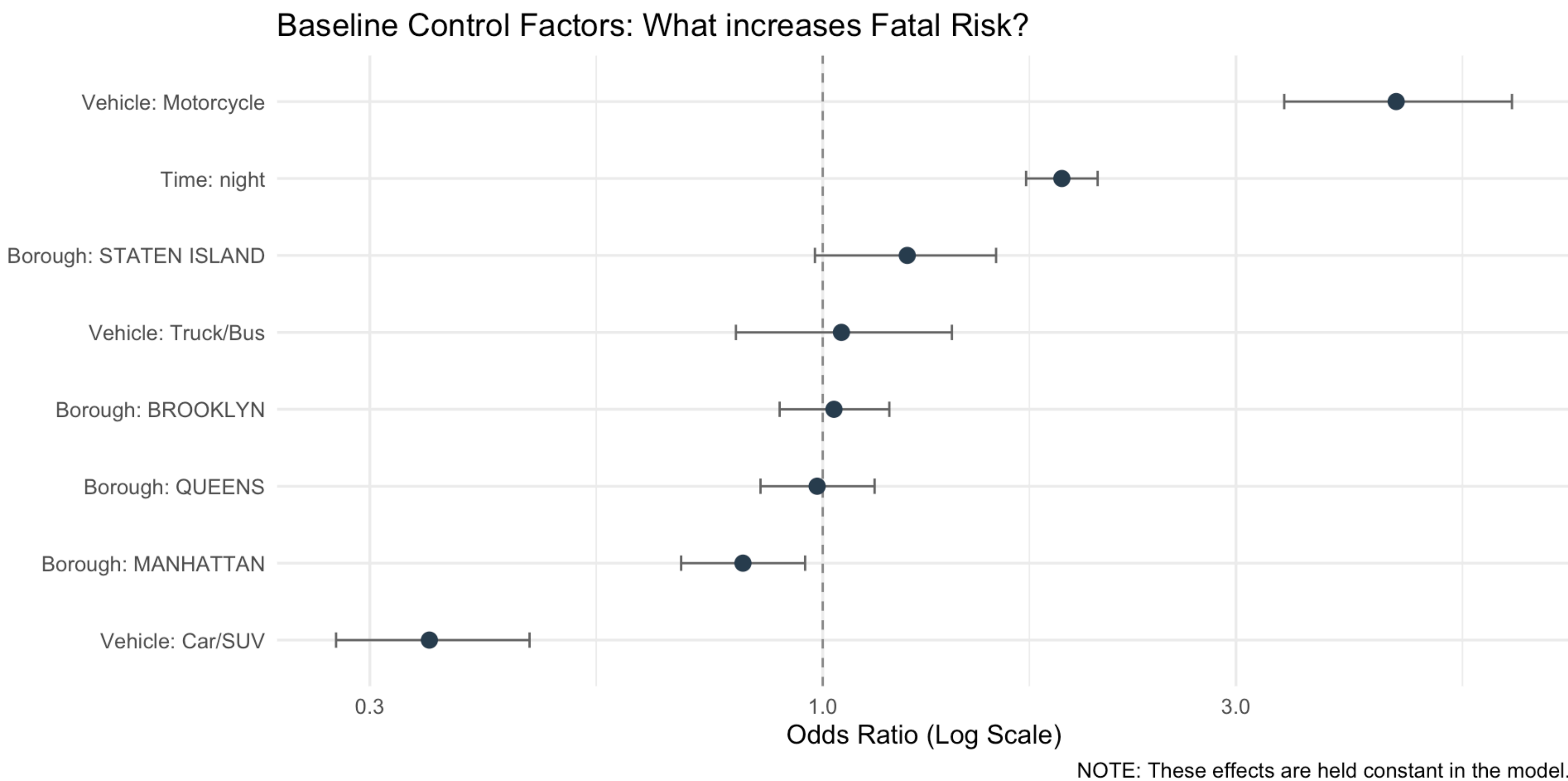
Top Phrases


Car Accidents on High-Speed Roads (>40 mph)
41.0°N
40.9°N
40.8°N
40.7°N
40.6°N
74.2°W
74.1°W
74.0°W
73.9°W
73.8°W
73.7°W
WHITE PLAINS
YONKERS
NEW ROCHELLE
PATERSON
PASSAIC
NEWARK
JERSEY CITY
NEW YORK
BAYONNE


Most Common Two Word Phrases in NYC Traffic Twee
cross bronx 2.6%
bronx expressway 2.1%
manhattan av 1.6%
exit 7n 1.6%
start route 1.4%
i87majordeeganexpressway sb 1.4%
bronxriverparkway sb 1.4%
trans manhattan 0.9%
tonnelle avenue 0.9%
river parkway 0.9%
queens kings 0.9%
parkway bronx 0.9%
manhattan expressway 0.9%
line koskuisko 0.9%
lane blocked 0.9%
kings county 0.9%
i278 wb 0.9%
fordham road 0.9%
0
1
2
3
Percentage of Total Phrases (%)

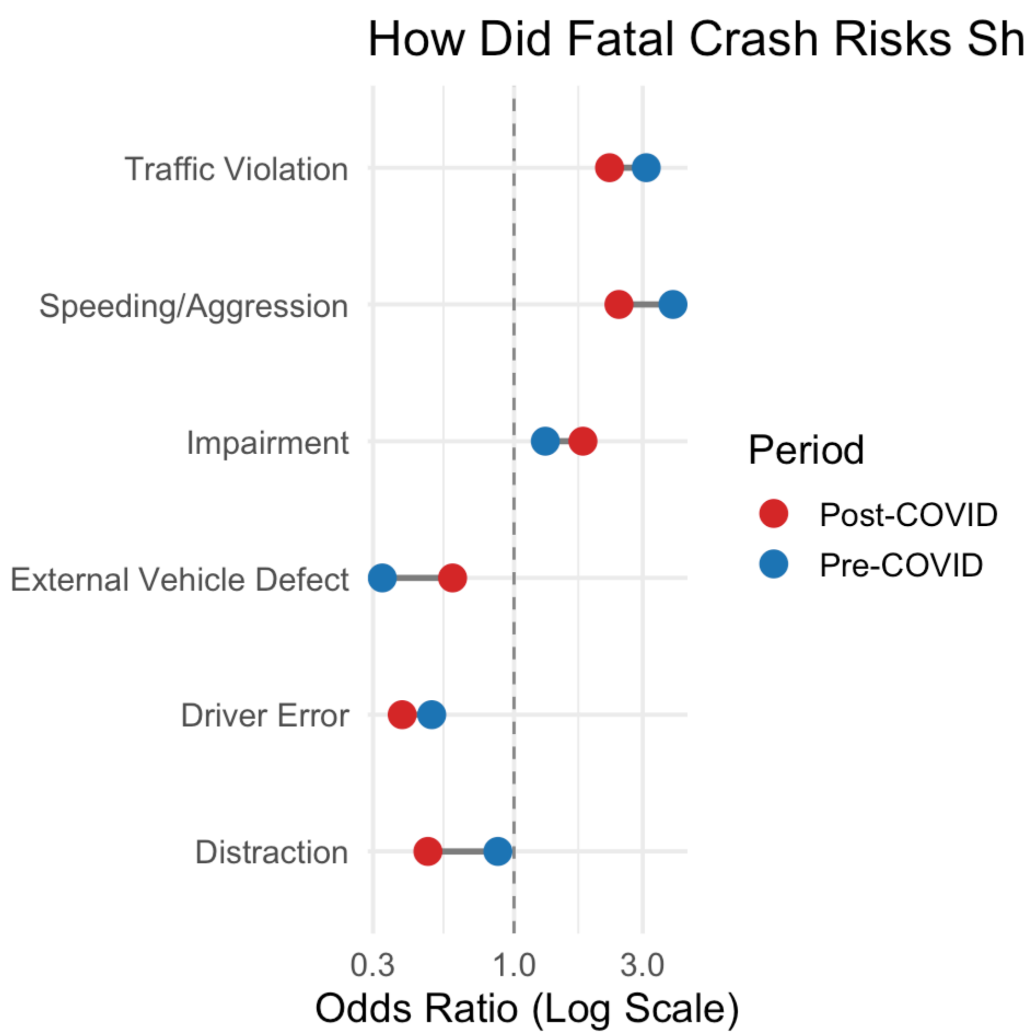
Statistical Modeling

Baseline Risk Factors



- **Motorcycles:** 3x+ more likely to be fatal
- **Nighttime:** ~2x higher odds than daytime
- **Manhattan:** Lower risk due to congestion

How COVID Changed Risk



Killer Plot

Interactive Killer Plot

Adjustable Map

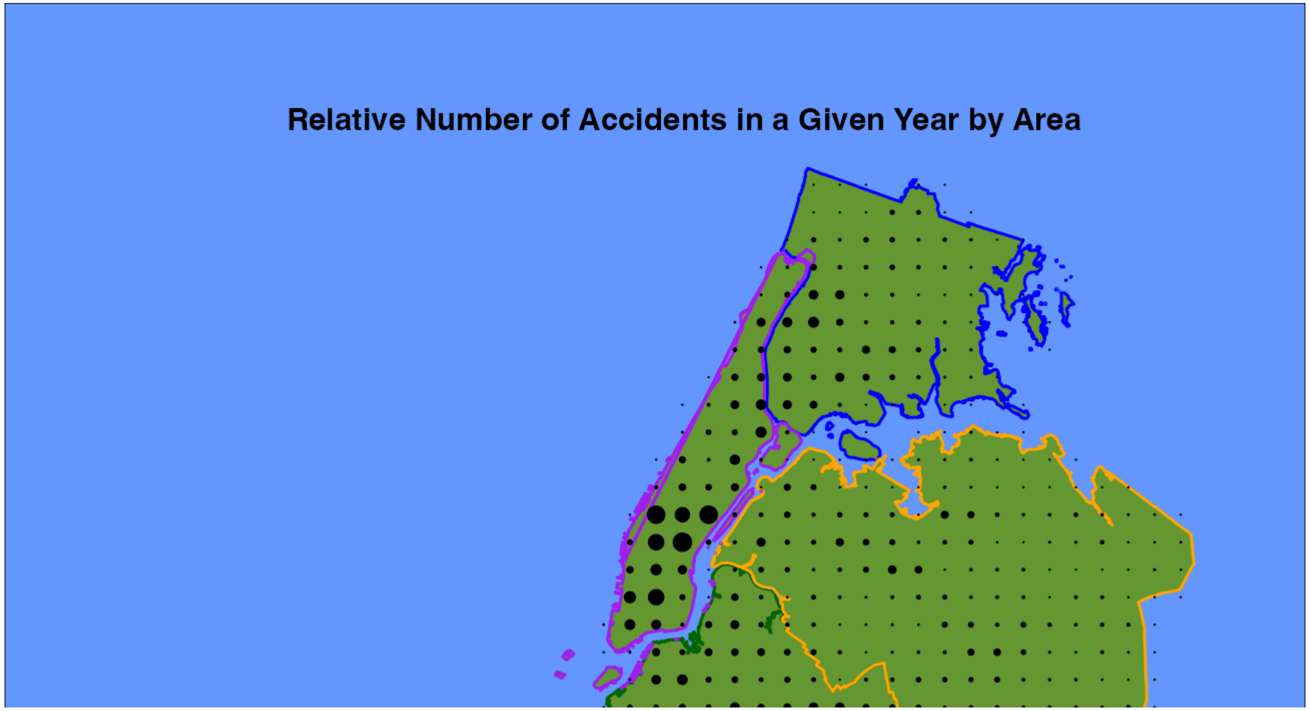
Choose view:

- (•) Accident Distribution
- ( ) Accident Trends

Year:

2,012 2,018 2,025

2,012 2,014 2,016 2,018 2,020 2,022 2,024



Conclusion

Key Findings

1. Collision trends
    - Rates stayed lower than before COVID and never fully returned
    - Borough distribution: Brooklyn and Queens dropped the most and recovered slowly; Manhattan rebounded fastest; the Bronx was intermediate
2. Severity trends
    - Fatality rate per crash rose sharply during and after COVID
    - Injuries per crash fell, indicating fewer but more lethal crashes
3. Driver-behavior
    - Speeding/Aggression and Impairment remained high-risk with unchanged odds
    - Distraction and Traffic-Violation odds fell significantly

Thank You!

Group 1: Astha Singh, Franco Bosetti, Louis Clarke, Lindsey Russ